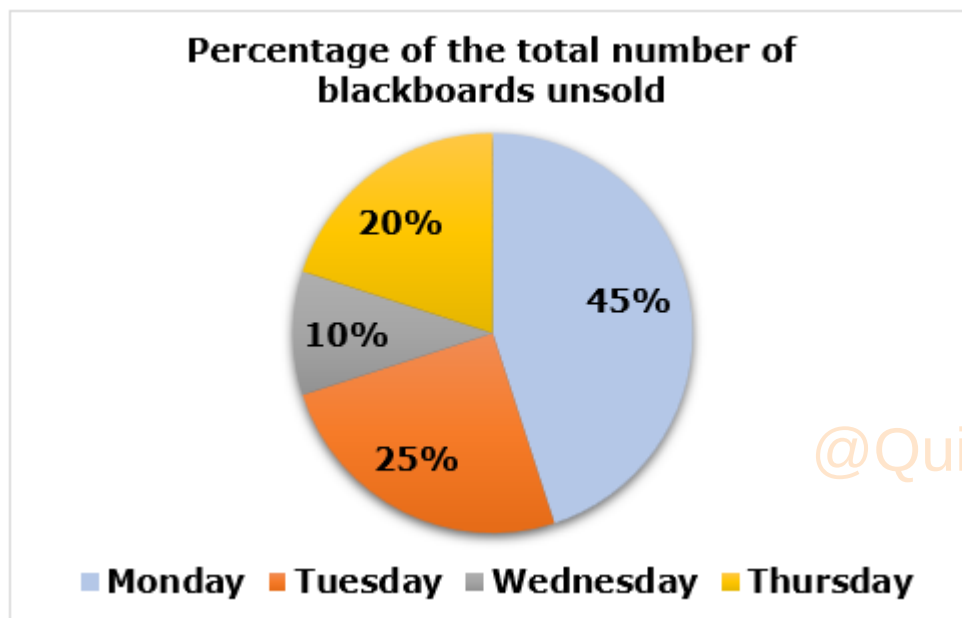
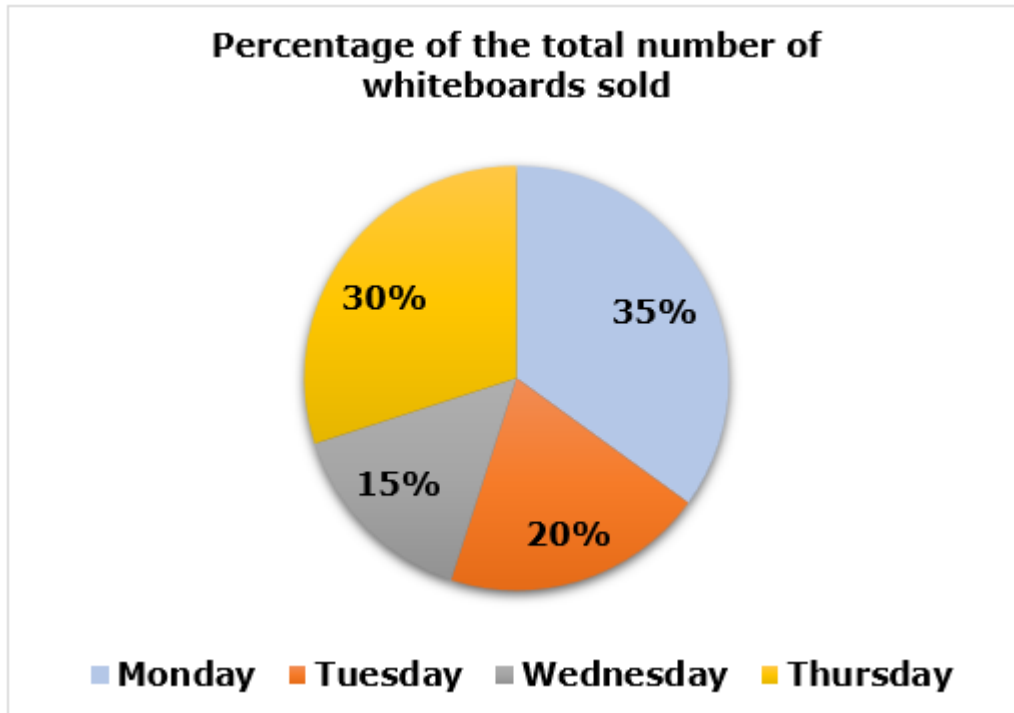


1. Questions

Read the following information carefully and answer the questions based on it.

The first pie chart given below shows the percentage distribution of the total number of whiteboards sold by a shop on four different days i.e., Monday, Tuesday, Wednesday and Thursday. The second pie chart given below shows the percentage distribution of the total number of blackboards unsold by the same shop on each day.



Note:

The total number of items manufactured = the total number of whiteboards manufactured (sold + unsold) + the total number of blackboards manufactured (sold + unsold)

The total number of items manufactured on Monday, Tuesday, Wednesday and Thursday is 440, 270, 200, 340 respectively.

The number of whiteboards unsold on Monday is 60, which is $\frac{2}{5}^{\text{th}}$ of the number of blackboards sold on the same day.

The number of blackboards sold on Tuesday is 100, which is 60 more than the number of whiteboards unsold on the same day.

On Wednesday, the number of whiteboards unsold is 20% of the number of blackboards sold.

On Thursday, the ratio of the number of blackboards sold and whiteboards unsold is 1:2.

Find the difference between the total number of whiteboards unsold and the total number of blackboards sold by the shop on all four days together.

- a. 147
- b. 152
- c. 186
- d. 170
- e. 135

2. Questions

On Monday, $\frac{1}{5}^{\text{th}}$ of the total number of whiteboards manufactured are steel whiteboards. If the ratio of the number of wooden and steel whiteboards sold on Monday is 11:3, then find the number of wooden whiteboards that remained unsold on Monday. (consider only wooden and steel whiteboards)

- a. 40
- b. 50
- c. 30
- d. 20
- e. 10

3. Questions

If the price of each blackboard on Thursday is 20% less than the price of each blackboard on Wednesday and the difference between the total revenue generated from the sale of all blackboards on Wednesday and Thursday is Rs. 26000, then find the price of each blackboard on Wednesday.

- a. Rs. 500
- b. Rs. 400
- c. Rs. 600
- d. Rs. 300
- e. Rs. 200

4. Questions

Find what percent of the total number of items (whiteboards + blackboards) manufactured by the shop on Tuesday are whiteboards?

- a. 33.33%
- b. 55.55%
- c. 44.44%
- d. 22.22%
- e. 11.11%

5. Questions

On Friday, if the total number of items manufactured by the shop is __[?]- less than the previous day and the total number of whiteboards manufactured on Friday is 200, which is twice the total number of blackboards manufactured on the same day.

What should be the correct replacement for the question mark?

- a. 24
- b. 56
- c. 62
- d. 38
- e. 40

6. Questions

Find the approximate value and determine what should come in place of the question mark?

$$44.88\% \text{ of } 1920.09 + ?^2 = 25.07\% \text{ of } 899.91 + 29.83\% \text{ of } 2250.03$$

- a. 8
- b. 3
- c. 9
- d. 5
- e. 6

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7. Questions

$$(34.93 * 5.13) - (24.08 * 3.86) = ? - (26.94 * 3.01) + (10.11 * 6.99)$$

- a. 72
- b. 90

- c. 65
- d. 53
- e. 38

8. Questions

$$(132.12 \div \sqrt{483.98}) + (\sqrt{1599.92} \div 8.06) - (240.13 \div 59.97) = \sqrt{?}$$

- a. 7
- b. 2401
- c. 65
- d. 49
- e. 38

9. Questions

$$(72.08\% \text{ of } 299.99) \div ? = (119.91\% \text{ of } 150.14) - (9.01\% \text{ of } 1899.85)$$

- a. 24
- b. 35
- c. 17
- d. 10
- e. 6

10. Questions

$$(35.92 * 10.08) + (70.07 * 1.98) - (19.91 * 11.02) + (45.04 * 3.96) = ?$$

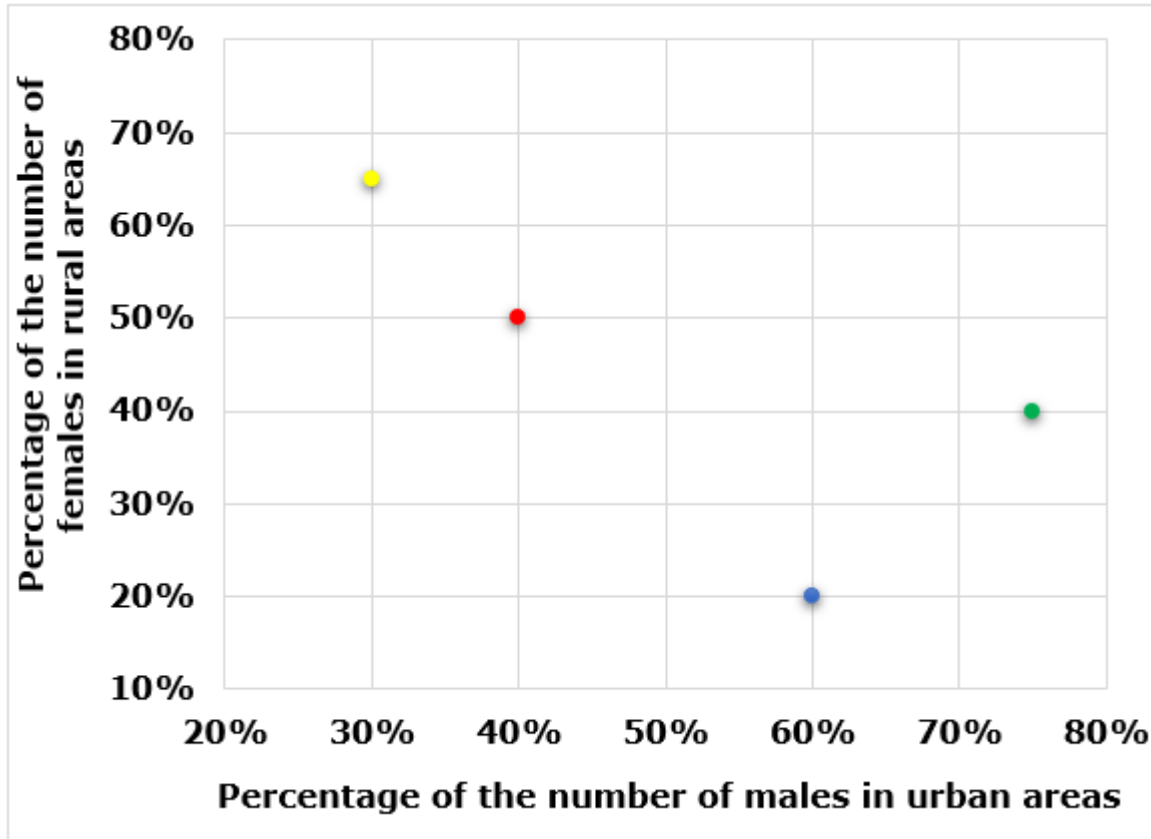
- a. 290
- b. 320
- c. 460
- d. 550
- e. 100

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11. Questions

Read the following information carefully and answer the questions based on it.

The scatter chart given below shows the percentage of the number of males in urban areas and the percentage of the number of females in rural areas in four different cities P, Q, R, and S.



Interpretation of figures: (X-Axis) represents the percentage of the number of males in urban areas and (Y-Axis) represents the percentage of the number of females in rural areas in each city. Blue, Red, Green, and Yellow data points represent cities P, Q, R, and S respectively.

Note:

The total population of any city = the total population (males + females) of urban areas + the total population (males + females) of rural areas

The total population (urban + rural) of cities P, Q, R, and S is 3500, 6500, 2000, and 5000 respectively.

In city P, the number of females in urban areas is $\frac{2}{3}$ rd of the number of males in rural areas.

The difference between the number of males in urban and rural areas in city Q is 800. (urban > rural)

The sum of the number of males in rural areas in city R and the number of females in urban areas in the same city is 780.

In city S, the number of females in rural areas is 2150 less than the number of females in urban areas.

Find the average of the total urban population of city P and the total rural population of city S.

- a. 1500
- b. 2000
- c. 1600
- d. 3200
- e. 4000

12. Questions

In the urban area of city Q, if the ratio of the number of married males and females is 1:2, while the number of unmarried females is 100 less than the number of unmarried males, then find the ratio of the number of married and unmarried males in urban area of city Q.

- a. 4:5
- b. 3:2
- c. 5:4
- d. 4:1
- e. 5:3

13. Questions

The number of females in the urban areas in city R is what percent more or less than the number of males in the rural areas in the same city?

- a. 37.50% more
- b. 37.50% less
- c. 62.50% less
- d. 87.50% more
- e. 12.50% less

14. Questions

In city P in rural areas, [X]% of the total population are illiterates and the ratio of the number of graduates and post-graduates is 3:1. If the number of post-graduates and the number of illiterates is the same, then find the value of $[X - 5]$. (Only literates can be graduates and post-graduates)

- a. 20
- b. 35
- c. 10
- d. 15
- e. 40

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15. Questions

In city S, if the number of children is also included, then the number of children in urban areas will be $\frac{1}{6}$ th of the total population of urban areas and the number of children in rural areas will be $\frac{1}{3}$ rd of the total population of rural areas. Find the total number of children in urban and rural areas in city S.

- a. 1300

- b. 2700
- c. 1200
- d. 1000
- e. 2000

16. Questions

The following question has two statements as statement I and statement II. You have to determine whether the data provided in two statements are sufficient to answer the question or not.

If the total distance between points A and B is 210 km, then find the original speed of the car.

Statement I: A car started from point A towards point B, but after 2 hours, it stopped somewhere between points A and B and covered the remaining distance with 10% more than its original speed in 2 hours.

Statement II: The time taken by the car to cover $\frac{2}{7}$ th of the total distance with 20% more than its original speed is 2 hours less than the time taken by the same car to cover the remaining distance with the original speed.

- a. Only statement I alone is sufficient
- b. Only statement II alone is sufficient
- c. Either statement I or II alone is sufficient
- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

17. Questions

Find the value of $[P^2]\%$ of 1600.

Statement I: An article whose cost price of Rs. $[24P]$ is marked-up by 50% above its cost price and sold at a discount of Rs. 20. The profit earned on selling the article is Rs. 40.

Statement II: The selling price of the article is Rs. $[32P]$, which sold at profit of 33.33%. The article was marked-up by Rs. 60.

- a. Only statement I alone is sufficient
- b. Only statement II alone is sufficient
- c. Either statement I or II alone is sufficient
- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

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18. Questions

There are two two-digit natural numbers AB and BA. Find the product of AB and BA.

Statement I: If AB is subtracted from BA, then the resultant will be a perfect cube of a single-digit prime number, which is less than 5 but more than 2.

Statement II: The sum of AB and BA is a two-digit natural number, whose both digits are the same, which is equal to the smallest multiple of 5.

- a. Only statement I alone is sufficient
- b. Only statement II alone is sufficient
- c. Either statement I or II alone is sufficient
- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

19. Questions

Find the difference between the amount invested in scheme M and N.

Statement I: The simple interest received from M at the rate of 30% per annum after 2 years is half of the amount invested in scheme N.

Statement II: The compound interest received from scheme N at the rate of 20% per annum compounded annually after 2 years is Rs. 180 less than the simple interest received from scheme M.

- a. Only statement I alone is sufficient
- b. Only statement II alone is sufficient
- c. Either statement I or II alone is sufficient
- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

20. Questions

Find the value of [D].

Statement I: The time taken by the boat to cover $[D + 13]$ km in downstream is 1 hour more than the time taken by the same boat to cover $[D - 20]$ km in upstream.

Statement II: The speed of the river is 3 km/hr, which is $\frac{1}{6}^{\text{th}}$ of the speed of the boat in still water.

- a. Only statement I alone is sufficient
- b. Only statement II alone is sufficient
- c. Either statement I or II alone is sufficient
- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

Explanations:

1. Questions

Let the total number of whiteboards sold on all four days together be $100m$.

Also, let the total number of blackboards unsold on all four days together be $100n$.

Monday and Tuesday

The number of whiteboards sold on Monday = 35% of $100m = 35m$

The number of whiteboards unsold on Monday = 60

The number of blackboards unsold on Monday = 45% of $100n = 45n$

The number of blackboards sold on Monday = $60/2 \times 5 = 150$

The total number of items manufactured on Monday = $(35m + 60 + 45n + 150) = (35m + 45n + 210)$

So,

$$35m + 45n + 210 = 440$$

$$35m + 45n = 230$$

$$7m + 9n = 46 \dots \dots \dots (I)$$

Now,

The number of whiteboards sold on Tuesday = 20% of $100m = 20m$

The number of blackboards sold on Tuesday = 100

The number of whiteboards unsold on Tuesday = $(100 - 60) = 40$

The number of blackboards unsold on Tuesday = 25% of $100n = 25n$

The total number of items manufactured on Monday = $(20m + 100 + 40 + 25n) = (20m + 25n + 140)$

So,

$$20m + 25n + 140 = 270$$

$$20m + 25n = 130$$

$$4m + 5n = 26 \dots \dots \dots (II)$$

By solving equation (I) and (II), we get,

$$28m + 36n = 184$$

$$28m + 35n = 182$$

$$n = 2$$

Now, put the value of n in equation (I), we get,

$$7m + 9n = 46$$

$$7m + 9 \times 2 = 46$$

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$$7m = 46 - 18$$

$$7m = 28$$

$$m = 4$$

So, the total number of whiteboards sold on all four days together = $100m = (100 * 4) = 400$

And, the total number of blackboards unsold on all four days together = $(100 * 2) = 200$

Monday

The number of whiteboards sold = $35m = (35 * 4) = 140$

The number of blackboards unsold = $45n = (45 * 2) = 90$

The number of whiteboards unsold = 60

The number of blackboards sold = 150

So, the total number of whiteboards manufactured = $(140 + 60) = 200$

And, the total number of blackboards manufactured = $(150 + 90) = 240$

Tuesday

The number of whiteboards sold = $20m = (20 * 4) = 80$

The number of blackboards unsold = $25n = (25 * 2) = 50$

The number of whiteboards unsold = 40

The number of blackboards sold = 100

So, the total number of whiteboards manufactured = $(80 + 40) = 120$

And, the total number of blackboards manufactured = $(100 + 50) = 150$

Wednesday

The total number of items manufactured = 200

The number of whiteboards sold = 15% of 400 = 60

The number of blackboards unsold = 10% of 200 = 20

The sum of the number of whiteboards unsold and blackboards sold = $200 - (60 + 20) = 200 - 80 = 120$

Also, the ratio of the number of whiteboards unsold and blackboards sold = $20:100 = 1:5$

So, the number of whiteboards unsold = $1/6$ of $120 = 20$

And, the number of blackboards sold = $5/6$ of $120 = 100$

So, the total number of whiteboards manufactured = $(60 + 20) = 80$

And, the total number of blackboards manufactured = $(100 + 20) = 120$

Thursday

The total number of items manufactured = 340

The number of whiteboards sold = 30% of 400 = 120

The number of blackboards unsold = 20% of 200 = 40

The sum of the number of whiteboards unsold and blackboards sold = $340 - (120 + 40) = 340 - 160 = 180$

So, the number of whiteboards unsold = $\frac{2}{3}$ of 180 = 120

And, the number of blackboards sold = $\frac{1}{3}$ of 180 = 60

So, the total number of whiteboards manufactured = $(120 + 120) = 240$

And, the total number of blackboards manufactured = $(60 + 40) = 100$

Whiteboards

Day	The total number of whiteboards manufactured	The number of whiteboards sold	The number of whiteboards unsold
Monday	200	140	60
Tuesday	120	80	40
Wednesday	80	60	20
Thursday	240	120	120
Total	640	400	240

Blackboards

Day	The total number of blackboards manufactured	The number of blackboards sold	The number of blackboards unsold
Monday	240	150	90
Tuesday	150	100	50
Wednesday	120	100	20
Thursday	100	60	40
Total	610	410	200

Answer: D

Now, according to the question,

The number of whiteboards unsold on,

Monday = 60

Tuesday = 40

Wednesday = 20

Thursday = 120

So, the total whiteboards unsold on all four days together = $(60 + 40 + 20 + 120) = 240$

The number of blackboards sold on,

Monday = 150

Tuesday = 100

Wednesday = 100

Thursday = 60

So, the total blackboards sold on all four days together = $(150 + 100 + 100 + 60) = 410$

Therefore, the required difference = $(410 - 240) = 170$.

Hence, the correct answer is option D.

2. Questions

Let the total number of whiteboards sold on all four days together be 100m.

Also, let the total number of blackboards unsold on all four days together be 100n.

Monday and Tuesday

The number of whiteboards sold on Monday = 35% of 100m = 35m

The number of whiteboards unsold on Monday = 60

The number of blackboards unsold on Monday = 45% of 100n = 45n

The number of blackboards sold on Monday = $60/2 \times 5 = 150$

The total number of items manufactured on Monday = $(35m + 60 + 45n + 150) = (35m + 45n + 210)$

So,

$$35m + 45n + 210 = 440$$

$$35m + 45n = 230$$

$$7m + 9n = 46 \dots \dots \dots (I)$$

Now,

The number of whiteboards sold on Tuesday = 20% of 100m = 20m

The number of blackboards sold on Tuesday = 100

The number of whiteboards unsold on Tuesday = $(100 - 60) = 40$

The number of blackboards unsold on Tuesday = 25% of 100n = 25n

The total number of items manufactured on Monday = $(20m + 100 + 40 + 25n) = (20m + 25n + 140)$

So,

$$20m + 25n + 140 = 270$$

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$$20m + 25n = 130$$

$$4m + 5n = 26 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$28m + 36n = 184$$

$$28m + 35n = 182$$

$$n = 2$$

Now, put the value of n in equation (I), we get,

$$7m + 9n = 46$$

$$7m + 9 * 2 = 46$$

$$7m = 46 - 18$$

$$7m = 28$$

$$m = 4$$

So, the total number of whiteboards sold on all four days together = $100m = (100 * 4) = 400$

And, the total number of blackboards unsold on all four days together = $(100 * 2) = 200$

Monday

The number of whiteboards sold = $35m = (35 * 4) = 140$

The number of blackboards unsold = $45n = (45 * 2) = 90$

The number of whiteboards unsold = 60

The number of blackboards sold = 150

So, the total number of whiteboards manufactured = $(140 + 60) = 200$

And, the total number of blackboards manufactured = $(150 + 90) = 240$

Tuesday

The number of whiteboards sold = $20m = (20 * 4) = 80$

The number of blackboards unsold = $25n = (25 * 2) = 50$

The number of whiteboards unsold = 40

The number of blackboards sold = 100

So, the total number of whiteboards manufactured = $(80 + 40) = 120$

And, the total number of blackboards manufactured = $(100 + 50) = 150$

Wednesday

The total number of items manufactured = 200

The number of whiteboards sold = 15% of 400 = 60

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The number of blackboards unsold = 10% of 200 = 20

The sum of the number of whiteboards unsold and blackboards sold = $200 - (60 + 20) = 200 - 80 = 120$

Also, the ratio of the number of whiteboards unsold and blackboards sold = 20:100 = 1:5

So, the number of whiteboards unsold = $1/6$ of 120 = 20

And, the number of blackboards sold = $5/6$ of 120 = 100

So, the total number of whiteboards manufactured = $(60 + 20) = 80$

And, the total number of blackboards manufactured = $(100 + 20) = 120$

Thursday

The total number of items manufactured = 340

The number of whiteboards sold = 30% of 400 = 120

The number of blackboards unsold = 20% of 200 = 40

The sum of the number of whiteboards unsold and blackboards sold = $340 - (120 + 40) = 340 - 160 = 180$

So, the number of whiteboards unsold = $2/3$ of 180 = 120

And, the number of blackboards sold = $1/3$ of 180 = 60

So, the total number of whiteboards manufactured = $(120 + 120) = 240$

And, the total number of blackboards manufactured = $(60 + 40) = 100$

Whiteboards

Day	The total number of whiteboards manufactured	The number of whiteboards sold	The number of whiteboards unsold
Monday	200	140	60
Tuesday	120	80	40
Wednesday	80	60	20
Thursday	240	120	120
Total	640	400	240

Blackboards

Day	The total number of blackboards manufactured	The number of blackboards sold	The number of blackboards unsold
Monday	240	150	90
Tuesday	150	100	50
Wednesday	120	100	20
Thursday	100	60	40
Total	610	410	200

Answer: B

Monday

The total number of whiteboards manufactured = 200

The number of steel whiteboards manufactured = $\frac{1}{5}$ of 200 = 40

The number of wooden whiteboards manufactured = $(200 - 40) = 160$

The total number of whiteboards sold = 140

The number of wooden whiteboards sold = $\frac{11}{14}$ of 140 = 110

The number of steel whiteboards sold = $\frac{3}{14}$ of 140 = 30

Therefore, the number of wooden whiteboards unsold on Monday = $(160 - 110) = 50$.

Hence, the correct answer is option B.

3. Questions

Let the total number of whiteboards sold on all four days together be 100m.

Also, let the total number of blackboards unsold on all four days together be 100n.

Monday and Tuesday

The number of whiteboards sold on Monday = 35% of 100m = 35m

The number of whiteboards unsold on Monday = 60

The number of blackboards unsold on Monday = 45% of 100n = 45n

The number of blackboards sold on Monday = $\frac{60}{2} \times 5 = 150$

The total number of items manufactured on Monday = $(35m + 60 + 45n + 150) = (35m + 45n + 210)$

So,

$$35m + 45n + 210 = 440$$

$$35m + 45n = 230$$

$$7m + 9n = 46 \dots \dots \dots (I)$$

Now,

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The number of whiteboards sold on Tuesday = 20% of 100m = 20m

The number of blackboards sold on Tuesday = 100

The number of whiteboards unsold on Tuesday = $(100 - 60) = 40$

The number of blackboards unsold on Tuesday = 25% of 100n = 25n

The total number of items manufactured on Monday = $(20m + 100 + 40 + 25n) = (20m + 25n + 140)$

So,

$$20m + 25n + 140 = 270$$

$$20m + 25n = 130$$

$$4m + 5n = 26 \dots \dots \dots (II)$$

By solving equation (I) and (II), we get,

$$28m + 36n = 184$$

$$28m + 35n = 182$$

$$n = 2$$

Now, put the value of n in equation (I), we get,

$$7m + 9n = 46$$

$$7m + 9 * 2 = 46$$

$$7m = 46 - 18$$

$$7m = 28$$

$$m = 4$$

So, the total number of whiteboards sold on all four days together = $100m = (100 * 4) = 400$

And, the total number of blackboards unsold on all four days together = $(100 * 2) = 200$

Monday

The number of whiteboards sold = $35m = (35 * 4) = 140$

The number of blackboards unsold = $45n = (45 * 2) = 90$

The number of whiteboards unsold = 60

The number of blackboards sold = 150

So, the total number of whiteboards manufactured = $(140 + 60) = 200$

And, the total number of blackboards manufactured = $(150 + 90) = 240$

Tuesday

The number of whiteboards sold = $20m = (20 * 4) = 80$

The number of blackboards unsold = $25n = (25 * 2) = 50$

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The number of whiteboards unsold = 40

The number of blackboards sold = 100

So, the total number of whiteboards manufactured = $(80 + 40) = 120$

And, the total number of blackboards manufactured = $(100 + 50) = 150$

Wednesday

The total number of items manufactured = 200

The number of whiteboards sold = 15% of 400 = 60

The number of blackboards unsold = 10% of 200 = 20

The sum of the number of whiteboards unsold and blackboards sold = $200 - (60 + 20) = 200 - 80 = 120$

Also, the ratio of the number of whiteboards unsold and blackboards sold = $20:100 = 1:5$

So, the number of whiteboards unsold = $1/6$ of 120 = 20

And, the number of blackboards sold = $5/6$ of 120 = 100

So, the total number of whiteboards manufactured = $(60 + 20) = 80$

And, the total number of blackboards manufactured = $(100 + 20) = 120$

Thursday

The total number of items manufactured = 340

The number of whiteboards sold = 30% of 400 = 120

The number of blackboards unsold = 20% of 200 = 40

The sum of the number of whiteboards unsold and blackboards sold = $340 - (120 + 40) = 340 - 160 = 180$

So, the number of whiteboards unsold = $2/3$ of 180 = 120

And, the number of blackboards sold = $1/3$ of 180 = 60

So, the total number of whiteboards manufactured = $(120 + 120) = 240$

And, the total number of blackboards manufactured = $(60 + 40) = 100$

Whiteboards

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Day	The total number of whiteboards manufactured	The number of whiteboards sold	The number of whiteboards unsold
Monday	200	140	60
Tuesday	120	80	40
Wednesday	80	60	20
Thursday	240	120	120
Total	640	400	240

Blackboards

Day	The total number of blackboards manufactured	The number of blackboards sold	The number of blackboards unsold
Monday	240	150	90
Tuesday	150	100	50
Wednesday	120	100	20
Thursday	100	60	40
Total	610	410	200

Answer: A

The ratio of the price of each blackboard on Wednesday and Thursday = $100:80 = 5:4$. Let it be Rs. 5k and 4k respectively.

The total number of blackboards sold on Wednesday = 100

The total number of blackboards sold on Thursday = 60

So, the total revenue generated from the sale of all blackboards on Wednesday = $(100 * 5k) = \text{Rs. } 500k$

And, the total revenue generated from the sale of all blackboards on Thursday = $(60 * 4k) = \text{Rs. } 240k$

Now,

$$500k - 240k = 26000$$

$$260k = 26000$$

$$k = 100$$

Therefore, the price of each blackboard on Wednesday = $5k = (5 * 100) = \text{Rs. } 500$.

Hence, the correct answer is option A.

4. Questions

Let the total number of whiteboards sold on all four days together be 100m.

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Also, let the total number of blackboards unsold on all four days together be 100n.

Monday and Tuesday

The number of whiteboards sold on Monday = 35% of 100m = 35m

The number of whiteboards unsold on Monday = 60

The number of blackboards unsold on Monday = 45% of 100n = 45n

The number of blackboards sold on Monday = $60/2 \times 5 = 150$

The total number of items manufactured on Monday = $(35m + 60 + 45n + 150) = (35m + 45n + 210)$

So,

$$35m + 45n + 210 = 440$$

$$35m + 45n = 230$$

$$7m + 9n = 46 \dots\dots\dots(I)$$

Now,

The number of whiteboards sold on Tuesday = 20% of 100m = 20m

The number of blackboards sold on Tuesday = 100

The number of whiteboards unsold on Tuesday = $(100 - 60) = 40$

The number of blackboards unsold on Tuesday = 25% of 100n = 25n

The total number of items manufactured on Monday = $(20m + 100 + 40 + 25n) = (20m + 25n + 140)$

So,

$$20m + 25n + 140 = 270$$

$$20m + 25n = 130$$

$$4m + 5n = 26 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$28m + 36n = 184$$

$$28m + 35n = 182$$

$$n = 2$$

Now, put the value of n in equation (I), we get,

$$7m + 9n = 46$$

$$7m + 9 \times 2 = 46$$

$$7m = 46 - 18$$

$$7m = 28$$

$$m = 4$$

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So, the total number of whiteboards sold on all four days together = $100m = (100 * 4) = 400$

And, the total number of blackboards unsold on all four days together = $(100 * 2) = 200$

Monday

The number of whiteboards sold = $35m = (35 * 4) = 140$

The number of blackboards unsold = $45n = (45 * 2) = 90$

The number of whiteboards unsold = 60

The number of blackboards sold = 150

So, the total number of whiteboards manufactured = $(140 + 60) = 200$

And, the total number of blackboards manufactured = $(150 + 90) = 240$

Tuesday

The number of whiteboards sold = $20m = (20 * 4) = 80$

The number of blackboards unsold = $25n = (25 * 2) = 50$

The number of whiteboards unsold = 40

The number of blackboards sold = 100

So, the total number of whiteboards manufactured = $(80 + 40) = 120$

And, the total number of blackboards manufactured = $(100 + 50) = 150$

Wednesday

The total number of items manufactured = 200

The number of whiteboards sold = 15% of 400 = 60

The number of blackboards unsold = 10% of 200 = 20

The sum of the number of whiteboards unsold and blackboards sold = $200 - (60 + 20) = 200 - 80 = 120$

Also, the ratio of the number of whiteboards unsold and blackboards sold = $20:100 = 1:5$

So, the number of whiteboards unsold = $1/6$ of 120 = 20

And, the number of blackboards sold = $5/6$ of 120 = 100

So, the total number of whiteboards manufactured = $(60 + 20) = 80$

And, the total number of blackboards manufactured = $(100 + 20) = 120$

Thursday

The total number of items manufactured = 340

The number of whiteboards sold = 30% of 400 = 120

The number of blackboards unsold = 20% of 200 = 40

The sum of the number of whiteboards unsold and blackboards sold = $340 - (120 + 40) = 340 - 160 = 180$

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So, the number of whiteboards unsold = $\frac{2}{3}$ of 180 = 120

And, the number of blackboards sold = $\frac{1}{3}$ of 180 = 60

So, the total number of whiteboards manufactured = $(120 + 120) = 240$

And, the total number of blackboards manufactured = $(60 + 40) = 100$

Whiteboards

Day	The total number of whiteboards manufactured	The number of whiteboards sold	The number of whiteboards unsold
Monday	200	140	60
Tuesday	120	80	40
Wednesday	80	60	20
Thursday	240	120	120
Total	640	400	240

Blackboards

Day	The total number of blackboards manufactured	The number of blackboards sold	The number of blackboards unsold
Monday	240	150	90
Tuesday	150	100	50
Wednesday	120	100	20
Thursday	100	60	40
Total	610	410	200

Answer: C

The total number of items manufactured on Tuesday = 270

The total number of whiteboards manufactured on Tuesday = 120

Therefore, the required percentage = $\frac{120}{270} \times 100 = 44.44\%$.

Hence, the correct answer is option C.

5. Questions

Let the total number of whiteboards sold on all four days together be 100m.

Also, let the total number of blackboards unsold on all four days together be 100n.

Monday and Tuesday

The number of whiteboards sold on Monday = 35% of 100m = 35m

The number of whiteboards unsold on Monday = 60

The number of blackboards unsold on Monday = 45% of 100n = 45n

The number of blackboards sold on Monday = $60/2 \times 5 = 150$

The total number of items manufactured on Monday = $(35m + 60 + 45n + 150) = (35m + 45n + 210)$

So,

$$35m + 45n + 210 = 440$$

$$35m + 45n = 230$$

$$7m + 9n = 46 \dots\dots\dots (I)$$

Now,

The number of whiteboards sold on Tuesday = 20% of 100m = 20m

The number of blackboards sold on Tuesday = 100

The number of whiteboards unsold on Tuesday = $(100 - 60) = 40$

The number of blackboards unsold on Tuesday = 25% of 100n = 25n

The total number of items manufactured on Monday = $(20m + 100 + 40 + 25n) = (20m + 25n + 140)$

So,

$$20m + 25n + 140 = 270$$

$$20m + 25n = 130$$

$$4m + 5n = 26 \dots\dots\dots (II)$$

By solving equation (I) and (II), we get,

$$28m + 36n = 184$$

$$28m + 35n = 182$$

$$n = 2$$

Now, put the value of n in equation (I), we get,

$$7m + 9n = 46$$

$$7m + 9 \times 2 = 46$$

$$7m = 46 - 18$$

$$7m = 28$$

$$m = 4$$

So, the total number of whiteboards sold on all four days together = $100m = (100 \times 4) = 400$

And, the total number of blackboards unsold on all four days together = $(100 \times 2) = 200$

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Monday

The number of whiteboards sold = $35m = (35 * 4) = 140$

The number of blackboards unsold = $45n = (45 * 2) = 90$

The number of whiteboards unsold = 60

The number of blackboards sold = 150

So, the total number of whiteboards manufactured = $(140 + 60) = 200$

And, the total number of blackboards manufactured = $(150 + 90) = 240$

Tuesday

The number of whiteboards sold = $20m = (20 * 4) = 80$

The number of blackboards unsold = $25n = (25 * 2) = 50$

The number of whiteboards unsold = 40

The number of blackboards sold = 100

So, the total number of whiteboards manufactured = $(80 + 40) = 120$

And, the total number of blackboards manufactured = $(100 + 50) = 150$

Wednesday

The total number of items manufactured = 200

The number of whiteboards sold = 15% of 400 = 60

The number of blackboards unsold = 10% of 200 = 20

The sum of the number of whiteboards unsold and blackboards sold = $200 - (60 + 20) = 200 - 80 = 120$

Also, the ratio of the number of whiteboards unsold and blackboards sold = $20:100 = 1:5$

So, the number of whiteboards unsold = $1/6$ of 120 = 20

And, the number of blackboards sold = $5/6$ of 120 = 100

So, the total number of whiteboards manufactured = $(60 + 20) = 80$

And, the total number of blackboards manufactured = $(100 + 20) = 120$

Thursday

The total number of items manufactured = 340

The number of whiteboards sold = 30% of 400 = 120

The number of blackboards unsold = 20% of 200 = 40

The sum of the number of whiteboards unsold and blackboards sold = $340 - (120 + 40) = 340 - 160 = 180$

So, the number of whiteboards unsold = $2/3$ of 180 = 120

And, the number of blackboards sold = $1/3$ of 180 = 60

@QuizByMickey

So, the total number of whiteboards manufactured = $(120 + 120) = 240$

And, the total number of blackboards manufactured = $(60 + 40) = 100$

Whiteboards

Day	The total number of whiteboards manufactured	The number of whiteboards sold	The number of whiteboards unsold
Monday	200	140	60
Tuesday	120	80	40
Wednesday	80	60	20
Thursday	240	120	120
Total	640	400	240

Blackboards

Day	The total number of blackboards manufactured	The number of blackboards sold	The number of blackboards unsold
Monday	240	150	90
Tuesday	150	100	50
Wednesday	120	100	20
Thursday	100	60	40
Total	610	410	200

Answer: E

The total number of whiteboards manufactured on Friday = 200

The total number of blackboards manufactured on Friday = $(200/2) = 100$

So, the total number of items manufactured on Friday = $(200 + 100) = 300$.

The total number of items manufactured on Thursday = 340

Therefore, the value of the question mark = $(340 - 300) = 40$.

Hence, the correct answer is option E.

6. Questions

Answer: E

$$44.88\% \text{ of } 1920.09 + ?^2 = 25.07\% \text{ of } 899.91 + 29.83\% \text{ of } 2250.03$$

$$45\% \text{ of } 1920 + ?^2 = 25\% \text{ of } 900 + 30\% \text{ of } 2250$$

$$864 + ?^2 = 225 + 675$$

$$864 + ?^2 = 900$$

$$?^2 = 900 - 864$$

$$?^2 = 36$$

$$? = \sqrt{36}$$

$$? = 6$$

Hence, the correct answer is option E.

7. Questions

Answer: B

$$(34.93 * 5.13) - (24.08 * 3.86) = ? - (26.94 * 3.01) + (10.11 * 6.99)$$

$$(35 * 5) - (24 * 4) = ? - (27 * 3) + (10 * 7)$$

$$175 - 96 = ? - 81 + 70$$

$$79 = ? - 11$$

$$79 + 11 = ?$$

$$90 = ?$$

Hence, the correct answer is option B.

8. Questions

Answer: D

$$(132.12 \div \sqrt{483.98}) + (\sqrt{1599.92} \div 8.06) - (240.13 \div 59.97) = \sqrt{?}$$

$$(132 \div \sqrt{484}) + (\sqrt{1600} \div 8) - (240 \div 60) = \sqrt{?}$$

$$(132 \div 22) + (40 \div 8) - (240 \div 60) = \sqrt{?}$$

$$6 + 5 - 4 = \sqrt{?}$$

$$11 - 4 = \sqrt{?}$$

$$7 = \sqrt{?}$$

$$7^2 = ?$$

$$49 = ?$$

Hence, the correct answer is option D.

9. Questions

Answer: A

$$(72.08\% \text{ of } 299.99) \div ? = (119.91\% \text{ of } 150.14) - (9.01\% \text{ of } 1899.85)$$

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$$(72\% \text{ of } 300) \div ? = (120\% \text{ of } 150) - (9\% \text{ of } 1900)$$

$$216 \div ? = 180 - 171$$

$$216 \div ? = 9$$

$$216 \div 9 = ?$$

$$24 = ?$$

Hence, the correct answer is option A.

10. Questions

Answer: C

$$(35.92 * 10.08) + (70.07 * 1.98) - (19.91 * 11.02) + (45.04 * 3.96) = ?$$

$$(36 * 10) + (70 * 2) - (20 * 11) + (45 * 4) = ?$$

$$360 + 140 - 220 + 180 = ?$$

$$500 - 40 = ?$$

$$460 = ?$$

Hence, the correct answer is option C.

11. Questions

City P

Let the total population of the urban area be a.

So, the total population of the rural area = $(3500 - a)$

The number of males in urban area = 60% of a = $0.60a$

The number of females in urban area = $(a - 0.60a) = 0.40a$

The number of females in rural area = 20% of $(3500 - a) = (700 - 0.20a)$

The number of males in rural area = $(3500 - a) - (700 - 0.20a) = (2800 - 0.80a)$

Now,

$$0.40a / (2800 - 0.80a) = 2/3$$

$$1.20a = 5600 - 1.60a$$

$$2.80a = 5600$$

$$a = 2000$$

Therefore,

The number of males in urban area = $0.60a = (0.60 * 2000) = 1200$

The number of females in urban area = $0.40a = (0.40 * 2000) = 800$

The total population of urban area = $(1200 + 800) = 2000$

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The number of females in rural area = $(700 - 0.20a) = (700 - 0.20 * 2000) = (700 - 400) = 300$

The number of males in rural area = $(2800 - 0.80a) = (2800 - 0.80 * 2000) = (2800 - 1600) = 1200$

The total population of the rural area = $(1200 + 300) = 1500$

City Q

Let the total population of urban areas be b .

So, the total population of the rural area = $(6500 - b)$

The number of males in urban area = 40% of $b = 0.40b$

The number of females in urban area = $(b - 0.40b) = 0.60b$

The number of females in rural area = 50% of $(6500 - b) = (3250 - 0.50b)$

The number of males in rural areas = $(6500 - b) - (3250 - 0.50b) = (3250 - 0.50b)$

Now,

$$0.40b - (3250 - 0.50b) = 800$$

$$0.90b = 4050$$

$$b = 4500$$

Therefore,

The number of males in urban area = $0.40b = (0.40 * 4500) = 1800$

The number of females in urban area = $0.60b = (0.60 * 4500) = 2700$

The total population of urban area = $(1800 + 2700) = 4500$

The number of females in rural area = $(3250 - 0.50b) = (3250 - 0.50 * 4500) = (3250 - 2250) = 1000$

The number of males in rural area = $(3250 - 0.50b) = (3250 - 2250) = 1000$

The total population of rural area = $(1000 + 1000) = 2000$

City R

Let the total population of the urban area be c .

So, the total population of the rural area = $(2000 - c)$

The number of males in urban area = 75% of $c = 0.75c$

The number of females in urban area = $(c - 0.75c) = 0.25c$

The number of females in rural area = 40% of $(2000 - c) = (800 - 0.40c)$

The number of males in rural area = $(2000 - c) - (800 - 0.40c) = (1200 - 0.60c)$

Now,

$$0.25c + (1200 - 0.60c) = 780$$

$$1200 - 0.35c = 780$$

@QuizByMickey

$$0.35c = 420$$

$$c = 1200$$

Therefore,

$$\text{The number of males in urban area} = 0.75c = (0.75 * 1200) = 900$$

$$\text{The number of females in urban area} = 0.25c = (0.25 * 1200) = 300$$

$$\text{The total population of urban area} = (900 + 300) = 1200$$

$$\text{The number of females in rural area} = (800 - 0.40c) = (800 - 0.40 * 1200) = (800 - 480) = 320$$

$$\text{The number of males in rural area} = (1200 - 0.60c) = (1200 - 0.60 * 1200) = (1200 - 720) = 480$$

$$\text{The total population of the rural area} = (480 + 320) = 800$$

City S

Let the total population of the urban area be d .

$$\text{So, the total population of the rural area} = (5000 - d)$$

$$\text{The number of males in urban area} = 30\% \text{ of } d = 0.30d$$

$$\text{The number of females in urban area} = (d - 0.30d) = 0.70d$$

$$\text{The number of females in rural area} = 65\% \text{ of } (5000 - d) = (3250 - 0.65d)$$

$$\text{The number of males in rural area} = (5000 - d) - (3250 - 0.65d) = (1750 - 0.35d)$$

Now,

$$0.70d - (3250 - 0.65d) = 2150$$

$$1.35d - 3250 = 2150$$

$$1.35d = 5400$$

$$d = 4000$$

Therefore,

$$\text{The number of males in urban area} = 0.30d = (0.30 * 4000) = 1200$$

$$\text{The number of females in urban area} = 0.70d = (0.70 * 4000) = 2800$$

$$\text{The total population of urban area} = (1200 + 2800) = 4000$$

$$\text{The number of females in rural area} = (3250 - 0.65d) = (3250 - 0.65 * 4000) = (3250 - 2600) = 650$$

$$\text{The number of males in rural area} = (1750 - 0.35d) = (1750 - 0.35 * 4000) = (1750 - 1400) = 350$$

$$\text{The total population of the rural area} = (350 + 650) = 1000$$

Urban Area

@QuizByMickey

City	The total population (males + females)	The number of males	The number of females
P	2000	1200	800
Q	4500	1800	2700
R	1200	900	300
S	4000	1200	2800
Total	11700	5100	6600

Rural Area

City	The total population (males + females)	The number of males	The number of females
P	1500	1200	300
Q	2000	1000	1000
R	800	480	320
S	1000	350	650
Total	5300	3030	2270

Answer: A

Now, according to the question,

The total urban population of city P = 2000

And, the total rural population of city S = 1000

Therefore, the required average = $(2000 + 1000) / 2 = 3000/2 = 1500$.

Hence, the correct answer is option A.

12. Questions

City P

Let the total population of the urban area be a.

So, the total population of the rural area = $(3500 - a)$

The number of males in urban area = 60% of a = $0.60a$

The number of females in urban area = $(a - 0.60a) = 0.40a$

The number of females in rural area = 20% of $(3500 - a) = (700 - 0.20a)$

The number of males in rural area = $(3500 - a) - (700 - 0.20a) = (2800 - 0.80a)$

Now,

$$0.40a / (2800 - 0.80a) = 2/3$$

$$1.20a = 5600 - 1.60a$$

@QuizByMickey

$$2.80a = 5600$$

$$a = 2000$$

Therefore,

$$\text{The number of males in urban area} = 0.60a = (0.60 * 2000) = 1200$$

$$\text{The number of females in urban area} = 0.40a = (0.40 * 2000) = 800$$

$$\text{The total population of urban area} = (1200 + 800) = 2000$$

$$\text{The number of females in rural area} = (700 - 0.20a) = (700 - 0.20 * 2000) = (700 - 400) = 300$$

$$\text{The number of males in rural area} = (2800 - 0.80a) = (2800 - 0.80 * 2000) = (2800 - 1600) = 1200$$

$$\text{The total population of the rural area} = (1200 + 300) = 1500$$

City Q

Let the total population of urban areas be b.

$$\text{So, the total population of the rural area} = (6500 - b)$$

$$\text{The number of males in urban area} = 40\% \text{ of } b = 0.40b$$

$$\text{The number of females in urban area} = (b - 0.40b) = 0.60b$$

$$\text{The number of females in rural area} = 50\% \text{ of } (6500 - b) = (3250 - 0.50b)$$

$$\text{The number of males in rural areas} = (6500 - b) - (3250 - 0.50b) = (3250 - 0.50b)$$

Now,

$$0.40b - (3250 - 0.50b) = 800$$

$$0.90b = 4050$$

$$b = 4500$$

Therefore,

$$\text{The number of males in urban area} = 0.40b = (0.40 * 4500) = 1800$$

$$\text{The number of females in urban area} = 0.60b = (0.60 * 4500) = 2700$$

$$\text{The total population of urban area} = (1800 + 2700) = 4500$$

$$\text{The number of females in rural area} = (3250 - 0.50b) = (3250 - 0.50 * 4500) = (3250 - 2250) = 1000$$

$$\text{The number of males in rural area} = (3250 - 0.50b) = (3250 - 2250) = 1000$$

$$\text{The total population of rural area} = (1000 + 1000) = 2000$$

City R

Let the total population of the urban area be c.

$$\text{So, the total population of the rural area} = (2000 - c)$$

$$\text{The number of males in urban area} = 75\% \text{ of } c = 0.75c$$

The number of females in urban area = $(c - 0.75c) = 0.25c$

The number of females in rural area = 40% of $(2000 - c) = (800 - 0.40c)$

The number of males in rural area = $(2000 - c) - (800 - 0.40c) = (1200 - 0.60c)$

Now,

$$0.25c + (1200 - 0.60c) = 780$$

$$1200 - 0.35c = 780$$

$$0.35c = 420$$

$$c = 1200$$

Therefore,

The number of males in urban area = $0.75c = (0.75 * 1200) = 900$

The number of females in urban area = $0.25c = (0.25 * 1200) = 300$

The total population of urban area = $(900 + 300) = 1200$

The number of females in rural area = $(800 - 0.40c) = (800 - 0.40 * 1200) = (800 - 480) = 320$

The number of males in rural area = $(1200 - 0.60c) = (1200 - 0.60 * 1200) = (1200 - 720) = 480$

The total population of the rural area = $(480 + 320) = 800$

City S

Let the total population of the urban area be d .

So, the total population of the rural area = $(5000 - d)$

The number of males in urban area = 30% of $d = 0.30d$

The number of females in urban area = $(d - 0.30d) = 0.70d$

The number of females in rural area = 65% of $(5000 - d) = (3250 - 0.65d)$

The number of males in rural area = $(5000 - d) - (3250 - 0.65d) = (1750 - 0.35d)$

Now,

$$0.70d - (3250 - 0.65d) = 2150$$

$$1.35d - 3250 = 2150$$

$$1.35d = 5400$$

$$d = 4000$$

Therefore,

The number of males in urban area = $0.30d = (0.30 * 4000) = 1200$

The number of females in urban area = $0.70d = (0.70 * 4000) = 2800$

The total population of urban area = $(1200 + 2800) = 4000$

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The number of females in rural area = $(3250 - 0.65d) = (3250 - 0.65 * 4000) = (3250 - 2600) = 650$

The number of males in rural area = $(1750 - 0.35d) = (1750 - 0.35 * 4000) = (1750 - 1400) = 350$

The total population of the rural area = $(350 + 650) = 1000$

Urban Area

City	The total population (males + females)	The number of males	The number of females
P	2000	1200	800
Q	4500	1800	2700
R	1200	900	300
S	4000	1200	2800
Total	11700	5100	6600

Rural Area

City	The total population (males + females)	The number of males	The number of females
P	1500	1200	300
Q	2000	1000	1000
R	800	480	320
S	1000	350	650
Total	5300	3030	2270

Answer: C

City Q (urban area)

The total number of males = 1800

The total number of females = 2700

Let the number of married males and females be 1:2. Let it be k and 2k respectively.

So, the number of unmarried males = $(1800 - k)$

And, the number of unmarried females = $(2700 - 2k)$

Now,

$$(1800 - k) - (2700 - 2k) = 100$$

$$1800 - k = 2700 - 2k$$

$$k = 1000$$

So, the number of married males = $k = 1000$

And, the number of unmarried males = $(1800 - k) = (1800 - 1000) = 800$

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Therefore, the required ratio = $1000:800 = 5:4$.

Hence, the correct answer is option C.

13. Questions

City P

Let the total population of the urban area be a .

So, the total population of the rural area = $(3500 - a)$

The number of males in urban area = 60% of $a = 0.60a$

The number of females in urban area = $(a - 0.60a) = 0.40a$

The number of females in rural area = 20% of $(3500 - a) = (700 - 0.20a)$

The number of males in rural area = $(3500 - a) - (700 - 0.20a) = (2800 - 0.80a)$

Now,

$$0.40a / (2800 - 0.80a) = 2/3$$

$$1.20a = 5600 - 1.60a$$

$$2.80a = 5600$$

$$a = 2000$$

Therefore,

$$\text{The number of males in urban area} = 0.60a = (0.60 * 2000) = 1200$$

$$\text{The number of females in urban area} = 0.40a = (0.40 * 2000) = 800$$

$$\text{The total population of urban area} = (1200 + 800) = 2000$$

$$\text{The number of females in rural area} = (700 - 0.20a) = (700 - 0.20 * 2000) = (700 - 400) = 300$$

$$\text{The number of males in rural area} = (2800 - 0.80a) = (2800 - 0.80 * 2000) = (2800 - 1600) = 1200$$

$$\text{The total population of the rural area} = (1200 + 300) = 1500$$

City Q

Let the total population of urban areas be b .

So, the total population of the rural area = $(6500 - b)$

The number of males in urban area = 40% of $b = 0.40b$

The number of females in urban area = $(b - 0.40b) = 0.60b$

The number of females in rural area = 50% of $(6500 - b) = (3250 - 0.50b)$

The number of males in rural areas = $(6500 - b) - (3250 - 0.50b) = (3250 - 0.50b)$

Now,

$$0.40b - (3250 - 0.50b) = 800$$

@QuizByMickey

$$0.90b = 4050$$

$$b = 4500$$

Therefore,

$$\text{The number of males in urban area} = 0.40b = (0.40 * 4500) = 1800$$

$$\text{The number of females in urban area} = 0.60b = (0.60 * 4500) = 2700$$

$$\text{The total population of urban area} = (1800 + 2700) = 4500$$

$$\text{The number of females in rural area} = (3250 - 0.50b) = (3250 - 0.50 * 4500) = (3250 - 2250) = 1000$$

$$\text{The number of males in rural area} = (3250 - 0.50b) = (3250 - 2250) = 1000$$

$$\text{The total population of rural area} = (1000 + 1000) = 2000$$

City R

Let the total population of the urban area be c.

$$\text{So, the total population of the rural area} = (2000 - c)$$

$$\text{The number of males in urban area} = 75\% \text{ of } c = 0.75c$$

$$\text{The number of females in urban area} = (c - 0.75c) = 0.25c$$

$$\text{The number of females in rural area} = 40\% \text{ of } (2000 - c) = (800 - 0.40c)$$

$$\text{The number of males in rural area} = (2000 - c) - (800 - 0.40c) = (1200 - 0.60c)$$

Now,

$$0.25c + (1200 - 0.60c) = 780$$

$$1200 - 0.35c = 780$$

$$0.35c = 420$$

$$c = 1200$$

Therefore,

$$\text{The number of males in urban area} = 0.75c = (0.75 * 1200) = 900$$

$$\text{The number of females in urban area} = 0.25c = (0.25 * 1200) = 300$$

$$\text{The total population of urban area} = (900 + 300) = 1200$$

$$\text{The number of females in rural area} = (800 - 0.40c) = (800 - 0.40 * 1200) = (800 - 480) = 320$$

$$\text{The number of males in rural area} = (1200 - 0.60c) = (1200 - 0.60 * 1200) = (1200 - 720) = 480$$

$$\text{The total population of the rural area} = (480 + 320) = 800$$

City S

Let the total population of the urban area be d.

$$\text{So, the total population of the rural area} = (5000 - d)$$

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The number of males in urban area = 30% of d = 0.30d

The number of females in urban area = (d - 0.30d) = 0.70d

The number of females in rural area = 65% of (5000 - d) = (3250 - 0.65d)

The number of males in rural area = (5000 - d) - (3250 - 0.65d) = (1750 - 0.35d)

Now,

$$0.70d - (3250 - 0.65d) = 2150$$

$$1.35d - 3250 = 2150$$

$$1.35d = 5400$$

$$d = 4000$$

Therefore,

$$\text{The number of males in urban area} = 0.30d = (0.30 * 4000) = 1200$$

$$\text{The number of females in urban area} = 0.70d = (0.70 * 4000) = 2800$$

$$\text{The total population of urban area} = (1200 + 2800) = 4000$$

$$\text{The number of females in rural area} = (3250 - 0.65d) = (3250 - 0.65 * 4000) = (3250 - 2600) = 650$$

$$\text{The number of males in rural area} = (1750 - 0.35d) = (1750 - 0.35 * 4000) = (1750 - 1400) = 350$$

$$\text{The total population of the rural area} = (350 + 650) = 1000$$

Urban Area

City	The total population (males + females)	The number of males	The number of females
P	2000	1200	800
Q	4500	1800	2700
R	1200	900	300
S	4000	1200	2800
Total	11700	5100	6600

Rural Area

City	The total population (males + females)	The number of males	The number of females
P	1500	1200	300
Q	2000	1000	1000
R	800	480	320
S	1000	350	650
Total	5300	3030	2270

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Answer: B

City R

The number of females in urban areas = 300

The number of males in rural areas = 480

So, the required percentage = $(480 - 300)/480 \times 100 = 180/480 \times 100 = 37.50\%$ less.

Hence, the correct answer is option B.

14. Questions

City P

Let the total population of the urban area be a .

So, the total population of the rural area = $(3500 - a)$

The number of males in urban area = 60% of $a = 0.60a$

The number of females in urban area = $(a - 0.60a) = 0.40a$

The number of females in rural area = 20% of $(3500 - a) = (700 - 0.20a)$

The number of males in rural area = $(3500 - a) - (700 - 0.20a) = (2800 - 0.80a)$

Now,

$$0.40a / (2800 - 0.80a) = 2/3$$

$$1.20a = 5600 - 1.60a$$

$$2.80a = 5600$$

$$a = 2000$$

Therefore,

$$\text{The number of males in urban area} = 0.60a = (0.60 \times 2000) = 1200$$

$$\text{The number of females in urban area} = 0.40a = (0.40 \times 2000) = 800$$

$$\text{The total population of urban area} = (1200 + 800) = 2000$$

$$\text{The number of females in rural area} = (700 - 0.20a) = (700 - 0.20 \times 2000) = (700 - 400) = 300$$

$$\text{The number of males in rural area} = (2800 - 0.80a) = (2800 - 0.80 \times 2000) = (2800 - 1600) = 1200$$

$$\text{The total population of the rural area} = (1200 + 300) = 1500$$

City Q

Let the total population of urban areas be b .

So, the total population of the rural area = $(6500 - b)$

The number of males in urban area = 40% of $b = 0.40b$

The number of females in urban area = $(b - 0.40b) = 0.60b$

The number of females in rural area = 50% of $(6500 - b) = (3250 - 0.50b)$

The number of males in rural areas = $(6500 - b) - (3250 - 0.50b) = (3250 - 0.50b)$

Now,

$$0.40b - (3250 - 0.50b) = 800$$

$$0.90b = 4050$$

$$b = 4500$$

Therefore,

$$\text{The number of males in urban area} = 0.40b = (0.40 * 4500) = 1800$$

$$\text{The number of females in urban area} = 0.60b = (0.60 * 4500) = 2700$$

$$\text{The total population of urban area} = (1800 + 2700) = 4500$$

$$\text{The number of females in rural area} = (3250 - 0.50b) = (3250 - 0.50 * 4500) = (3250 - 2250) = 1000$$

$$\text{The number of males in rural area} = (3250 - 0.50b) = (3250 - 2250) = 1000$$

$$\text{The total population of rural area} = (1000 + 1000) = 2000$$

City R

Let the total population of the urban area be c .

$$\text{So, the total population of the rural area} = (2000 - c)$$

$$\text{The number of males in urban area} = 75\% \text{ of } c = 0.75c$$

$$\text{The number of females in urban area} = (c - 0.75c) = 0.25c$$

$$\text{The number of females in rural area} = 40\% \text{ of } (2000 - c) = (800 - 0.40c)$$

$$\text{The number of males in rural area} = (2000 - c) - (800 - 0.40c) = (1200 - 0.60c)$$

Now,

$$0.25c + (1200 - 0.60c) = 780$$

$$1200 - 0.35c = 780$$

$$0.35c = 420$$

$$c = 1200$$

Therefore,

$$\text{The number of males in urban area} = 0.75c = (0.75 * 1200) = 900$$

$$\text{The number of females in urban area} = 0.25c = (0.25 * 1200) = 300$$

$$\text{The total population of urban area} = (900 + 300) = 1200$$

$$\text{The number of females in rural area} = (800 - 0.40c) = (800 - 0.40 * 1200) = (800 - 480) = 320$$

$$\text{The number of males in rural area} = (1200 - 0.60c) = (1200 - 0.60 * 1200) = (1200 - 720) = 480$$

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The total population of the rural area = $(480 + 320) = 800$

City S

Let the total population of the urban area be d .

So, the total population of the rural area = $(5000 - d)$

The number of males in urban area = 30% of $d = 0.30d$

The number of females in urban area = $(d - 0.30d) = 0.70d$

The number of females in rural area = 65% of $(5000 - d) = (3250 - 0.65d)$

The number of males in rural area = $(5000 - d) - (3250 - 0.65d) = (1750 - 0.35d)$

Now,

$$0.70d - (3250 - 0.65d) = 2150$$

$$1.35d - 3250 = 2150$$

$$1.35d = 5400$$

$$d = 4000$$

Therefore,

The number of males in urban area = $0.30d = (0.30 * 4000) = 1200$

The number of females in urban area = $0.70d = (0.70 * 4000) = 2800$

The total population of urban area = $(1200 + 2800) = 4000$

The number of females in rural area = $(3250 - 0.65d) = (3250 - 0.65 * 4000) = (3250 - 2600) = 650$

The number of males in rural area = $(1750 - 0.35d) = (1750 - 0.35 * 4000) = (1750 - 1400) = 350$

The total population of the rural area = $(350 + 650) = 1000$

Urban Area

City	The total population (males + females)	The number of males	The number of females
P	2000	1200	800
Q	4500	1800	2700
R	1200	900	300
S	4000	1200	2800
Total	11700	5100	6600

Rural Area

City	The total population (males + females)	The number of males	The number of females
P	1500	1200	300
Q	2000	1000	1000
R	800	480	320
S	1000	350	650
Total	5300	3030	2270

Answer: D

City P (rural area)

The total population = 1500

The number of illiterates = $X\%$ of 1500 = $15X$

The number of literates = $(1500 - 15X)$

The number of post-graduates = $15X$

The number of graduates = $(1500 - 15X) - 15X = (1500 - 30X)$

Now,

$$(1500 - 30X) / 15X = 3/1$$

$$1500 - 30X = 45X$$

$$75X = 1500$$

$$X = 20$$

Therefore, $(X - 5) = (20 - 5) = 15$.

Hence, the correct answer is option D.

15. Questions

City P

Let the total population of the urban area be a .

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$$0.40a / (2800 - 0.80a) = 2/3$$

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Now,

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$$0.90b = 4050$$

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Therefore,

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$$\text{The total population of urban area} = (1800 + 2700) = 4500$$

$$\text{The number of females in rural area} = (3250 - 0.50b) = (3250 - 0.50 * 4500) = (3250 - 2250) = 1000$$

$$\text{The number of males in rural area} = (3250 - 0.50b) = (3250 - 2250) = 1000$$

$$\text{The total population of rural area} = (1000 + 1000) = 2000$$

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Let the total population of the urban area be c.

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@QuizByMickey

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The total population of urban area = $(1200 + 2800) = 4000$

The number of females in rural area = $(3250 - 0.65d) = (3250 - 0.65 * 4000) = (3250 - 2600) = 650$

The number of males in rural area = $(1750 - 0.35d) = (1750 - 0.35 * 4000) = (1750 - 1400) = 350$

The total population of the rural area = $(350 + 650) = 1000$

Urban Area

City	The total population (males + females)	The number of males	The number of females
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S	4000	1200	2800
Total	11700	5100	6600

Rural Area

City	The total population (males + females)	The number of males	The number of females
P	1500	1200	300
Q	2000	1000	1000
R	800	480	320
S	1000	350	650
Total	5300	3030	2270

Answer: A

City S

Urban Area

Let the number of children be y .

The total number of males and females = $(1200 + 2800) = 4000$

Now, the total population = $(4000 + y)$

So,

$$y / (4000 + y) = 1/6$$

$$6y = 4000 + y$$

$$5y = 4000$$

$$y = 800$$

So, the number of children in urban areas = $y = 800$

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Rural Area

Let the number of children be z .

The total number of males and females = $(350 + 650) = 1000$

Now, the total population = $(1000 + z)$

So,

$$z / (1000 + z) = 1/3$$

$$3z = 1000 + z$$

$$2z = 1000$$

$$z = 500$$

So, the number of children in rural areas = $z = 500$

Therefore, the total number of children (urban + rural) in city S = $(y + z) = (800 + 500) = 1300$.

Hence, the correct answer is option A.

16. Questions

Answer: C

The total distance between points A and B = 210 km.

Let the original speed of the car be V km/hr.

Using statement I alone

The distance covered by the car with original speed = $(2 * V) = 2V$ km

And, the remaining distance covered by the car with increased speed = $(110\% \text{ of } V) * 2 = (1.10V * 2) = 2.20V$ km

Now,

$$2V + 2.20V = 210$$

$$4.20V = 210$$

$$V = 50$$

So, the original speed of the car = $V = 50$ km/hr.

Statement I alone is sufficient.

Using statement II alone

The increased speed of the car = $120\% \text{ of } V = 1.20V$ km/hr

The distance covered by the car in the first part = $2/7 \text{ of } 210 = 60$ km

So, the remaining distance covered by the car in the second part = $(210 - 60) = 150$ km

Now,

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$$(150 / V) - (60 / 1.20V) = 2$$

$$(180V - 60V) / 1.20V^2 = 2$$

$$2.40V^2 = 120V$$

$$2.40V = 120$$

$$V = 50$$

So, the original speed of the car = $V = 50$ km/hr.

Statement II alone is sufficient.

Therefore, either statement I or II alone is sufficient.

Hence, the correct answer is option C.

17. Questions

Answer: A

Using statement I alone

The C.P. = Rs. 24P

The M.P. = 150% of 24P = Rs. 36P

The S.P. = Rs. (36P - 20)

Now,

$$(36P - 20) - 24P = 40$$

$$12P = 60$$

$$P = 5$$

So, $P^2\%$ of 1600 = $5^2\%$ of 1600 = 25% of 1600 = 400.

Therefore, statement I alone is sufficient.

Using statement II alone

The ratio of C.P. and S.P. = 3:4. (Note: 33.33% = 1/3)

The S.P. = Rs. 32P

The C.P. = (32P/4)*3 = Rs. 24P

The M.P. = Rs. (24P + 60)

From here, we cannot find the value of P.

So, statement II alone is not sufficient.

Therefore, only statement I alone is sufficient.

Hence, the correct answer is option A.

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18. Questions

Answer: D

Using statement I and II together

The only single-digit prime number, which is less than 5 but more than 2 = 3.

So,

$$BA - AB = 3^3$$

$$BA - AB = 27$$

$$(10B+A) - (10A+B) = 27$$

$$9B - 9A = 27$$

$$B - A = 3 \dots \dots \dots (i)$$

Then the possible value of BA = 41, 52, 63, 74, 85 or 96

Now,

The smallest multiple of 5 = 5.

So,

$$BA + AB = 55$$

$$(10B+A) + (10A+B) = 55$$

$$11B + 11A = 55$$

$$B + A = 5 \dots \dots \dots (ii)$$

By solving equation (i) and (ii), we get,

$$A = 1$$

$$B = 4$$

Then the only possible value of BA = 41 and AB = 14

$$\text{So, } (AB * BA) = (14 * 41) = 574.$$

Therefore, both statements I and II together are necessary.

Hence, the correct answer is option D.

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19. Questions

Answer: D

Using statements I and II together

Let the amount invested in scheme M be Rs. 100x.

So, the simple interest received from scheme M = $(30 * 2)\%$ of 100x = 60% of 100x = Rs. 60x

Now, the amount invested in scheme N = $(60x * 2) = \text{Rs. } 120x$

The rate of interest for scheme N = $(20 + 20) + (20 * 20) / 100 = (40 + 4)\% = 44\%$.

So, the compound interest received from scheme N = 44% of $120x = \text{Rs. } 52.80x$

Now,

$$60x - 52.80x = 180$$

$$7.20x = 180$$

$$x = 25$$

So, the required difference = $(120x - 100x) = 20x = (20 * 25) = \text{Rs. } 500$.

Therefore, statements I and II together are necessary.

Hence, the correct answer is option D.

20. Questions

Answer: D

Using statements I and II together

The speed of the river = 3 km/hr

The speed of the boat in still water = $(3 * 6) = 18$ km/hr

The downstream speed = $(18 + 3) = 21$ km/hr

The upstream speed = $(18 - 3) = 15$ km/hr

Now,

$$(D + 13) / 21 - (D - 20) / 15 = 1$$

$$(5D + 65) - (7D - 140) / 105 = 1$$

$$205 - 2D = 105$$

$$2D = 100$$

$$D = 50$$

Therefore, both statements I and II together are necessary.

Hence, the correct answer is option D.

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